

Shibin Biji Thomas

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Summary

Software Engineer (Graduating May 2027) with hands-on experience designing, building, and scaling end-to-end web, platform, and data applications. Proficient in Java, Python, C++, JavaScript, TypeScript, and cloud-native infrastructure. Strong background across the full software development lifecycle (SDLC)—including architectural design, unit testing, automated CI/CD, debugging, and AI-assisted development tools (GitHub Copilot, LLM workflows).

Education

San Jose State University, San Jose, CA

Jan 2025 – May 2027

Master of Science in Applied Data Intelligence

GPA: 3.5

- **Coursework:** Data Structures & Algorithms, Operating Systems, Distributed Systems, Object-Oriented Design.

Skills

Languages: Java, Python, C++, JavaScript, TypeScript, C#, SQL, HTML5/CSS3, PyTorch, Vector Search

Web, Mobile & Platform Infrastructure: React, Node.js, RESTful APIs, Microservices, Scalable Systems, GraphQL

SDLC, Quality & Testing: Full SDLC, Unit Testing, Integration Testing, Debugging, Code Reviews, CI/CD Pipelines, Git

Datastores & Cloud: PostgreSQL, MongoDB, Redis, Docker, Kubernetes, AWS, GCP

AI-Assisted Tooling & Analytics: AI-Assisted Coding Tools (GitHub Copilot, AI Debugging), Performance Profiling

Experience

AI/ML Intern

Sunnyvale, CA, USA

LinkedIn

May 2026 – Aug 2026

- **Architected and benchmarked novel Multi-Task Learning (MTL) heads** (MMoE v3, 2-Level PLE, Grouped/Capped CGC, DCNv2) for a **12-objective feed ranking model** (m1486), systematically resolving **gradient conflict and capacity contention** across sparse and high-frequency engagement signals.
- **Delivered top-tier ranking quality gains**, achieving **11/12 AUC wins** (up to +0.20% Click, +0.34% Save, +1.00% Message) and **11/12 Normalized Entropy (NE) improvements** (up to -0.87% Vote NE, -0.82% Message NE) .
- **Designed and deployed a Hybrid Static-Head + Hashed-Tail embedding layer** for ultra-high cardinality features (hashedActorId, 201.5M entities across 79.5B training impressions), **eliminating severe hash collision interference** across 16.78M embedding rows.
- **Optimized multi-task gate routing** by implementing a 20% capped private capacity constraint, gate dropout (0.15), and learnable softmax scales, **preventing high-frequency signals** (*click, like*) **from monopolizing expert capacity**.
- **Implemented distributed GPU state synchronization** (All-Gather sync across PyTorch DDP ranks) for dynamic registry updates and **evaluated LFU eviction policies**, identifying key trade-offs between eviction.

Software Engineer

Pune, India

Accenture Pvt. Ltd.

July 2021 – Dec 2024

- Developed and deployed predictive machine learning models (**Scikit-learn, PyTorch, Python**) for customer classification and lifecycle churn prediction, leveraging complex **SQL** queries on large-scale relational datasets.
- Conducted data mining, statistical analysis, and automated feature selection on multi-terabyte customer datasets, improving predictive model precision and recall by **18%**.
- Containerized ML scoring microservices using **Docker** and deployed scalable inference pipelines on **AWS (ECS, S3)** with continuous integration (**CI/CD**), ensuring **99.6%** system uptime.
- Designed unit and integration testing suites (**PyTest, JUnit**) for automated ML data validation and API endpoints, reducing production data pipeline defects by **24%**.

Projects

AI-Driven Multi-Modal Personalization & Media Platform (Python, PyTorch, Scikit-learn, AWS, Docker)

- Architected a multi-modal machine learning platform in **Python (FastAPI, PyTorch)**, implementing deep learning feature extraction and media personalization algorithms on large-scale datasets.
- Integrated vector similarity search using high-dimensional embeddings and **Redis** caching, reducing media search and retrieval latency to sub-80ms across 100,000+ asset catalog entries.
- Applied statistical modeling and data mining methods in **Scikit-learn** to automate metadata tag extraction and customer engagement classification.
- Containerized ML serving infrastructure with **Docker** and configured **CI/CD** pipelines on **AWS (ECS, S3)**, utilizing asynchronous task queues (**Celery**) to guarantee **99.9%** service uptime.
- **Technologies Used:** Python, PyTorch, Scikit-learn, FastAPI, Redis, Docker, AWS (ECS, S3), Vector Search, SQL

Distributed Time Series Analytics & Event Processing Engine (Java, Python, C++, SQL, PostgreSQL)

- Engineered a real-time telemetry processing engine in **Java (Spring Boot)** and **C++**, ingesting and aggregating 15,000+ time-series user activity events per second for predictive analytics.
- Implemented advanced relational schema partitioning, indexing, and SQL optimization in **PostgreSQL**, decreasing query latency by **35%** under high write volumes.
- Developed statistical time-series forecasting scripts in **Python** to detect data drift and operational performance outliers across stream pipelines.